

TACTIX AI

AI-Assisted Simulated Mission-Planning & Risk Analysis Platform

Comprehensive Technical Architecture, Routing Algorithms, Datasets, and Workflow Guide

Executive Summary: TACTIX AI is a web-based decision-support platform designed to demonstrate how real environmental data (satellite terrain elevation, road infrastructure, satellite land-cover, and hourly weather reanalysis), synthetic operational data (logistics capacity, sensor uncertainty), deterministic mathematical simulation (A* multi-objective pathfinding and a 6-factor risk engine), and large language models (LLMs) combine into a single, explainable tool. It is built strictly as a controlled simulation prototype for fictional training scenarios.

1. Core Concepts & Principles Used in TACTIX AI

To understand how TACTIX AI operates, it helps to understand its core design pillars:

- **Explainability Over Black-Box AI:** Traditional AI route planners often use deep neural networks that output a path without explaining *why* that path was chosen. TACTIX AI uses a transparent, auditable weighted mathematical formula where every single point of risk can be traced directly to physical slope, soil moisture, weather friction, or fuel draw.
- **Deterministic Core, Optional AI Shell:** The pathfinder, physics simulation, constraint checker, and risk score engine are written in 100% pure TypeScript algorithms that run client-side. If the internet is disconnected or OpenAI goes down, the entire simulation engine continues to work perfectly.
- **Human-in-the-Loop Decision Support:** The AI acts as an advisor/analyst that translates words into simulation parameters and explains trade-offs. The AI **never** invents numerical risk scores, never overrides human judgment, and never picks a 'best military plan'—it presents evaluated alternatives for human evaluation.
- **Progressive Disclosure (Simple View vs Advanced View):** Designed for both beginners and deep technical analysts. By default, the interface presents simple, clean option cards with plain English explanations and [Why?] buttons. With one click, power users can expand full raw GIS telemetry.

2. Technologies & Programming Languages Used

The platform is architected across four clean layers using modern, production-grade tools:

Layer	Technologies / Libraries	Role in TACTIX AI
Frontend (UI)	Next.js 15 (App Router) React 19 TypeScript Tailwind CSS	Provides responsive layout, Tactical Ops command-center theme, interactive slic
Mapping & Charts	MapLibre GL JS HTML5 Canvas Recharts Lucide Icons	Renders 10 toggleable map layers (terrain, roads, water, weather, routes) with cl

Simulation Engine (Backend Core)	Pure TypeScript (zero external runtime dependencies)	Executes A* multi-objective pathfinding, environment cost evaluation, duration/fuel
AI & API Layer	Next.js Serverless Routes OpenAI GPT-4.1-mini Zod Schema Validator	<code><code>/api/interpret</code></code> translates natural language into structured parameters
GIS Data Pipeline & ML Experiment	Python 3.11 NumPy, Rasterio, Shapely GeoPandas, Scikit-learn	Offline GIS data clipping and raster processing. Trains Random Forest Regression

3. Backend Datasets & How They Are Processed

TACTIX AI fuses four real public environmental datasets with two synthetic operational feeds. All data is clipped to a shared 42.8 km² study area (Chennai Coastal & River Corridor, 12.95°N–13.01°N, 80.13°E–80.20°E):

Dataset	Source & Type	Variables Extracted	How It Affects Routing & Risk
1. SRTM DEM (Terrain)	USGS / NASA (Real 30m Global DEM)	Elevation, Slope (deg), Aspect, Roughness (std-dev)	Steep slopes and rocky roughness slow vehicle transit and increase fuel
2. OpenStreetMap (Infrastructure)	OSM Community (Real Vector Layers)	Roads, Waterways, Bridges, Buildings	Cells near paved roads get high accessibility bonuses (+speed, -fuel)
3. Copernicus (Land Cover)	European Union (Real Satellite Map)	Urban (0.5), Forest (0.7), Grassland (0.3), Water (1.0), Wetland (0.8), Bare (0.35)	Each land type has a friction multiplier. Open fields (0.3) are fast; forests (0.7) are slow
4. ERA5 Weather	ECMWF (Real Reanalysis)	Rain (mm), Wind (m/s), Pressure (hPa), Visibility	Classifies weather into 5 tiers (Normal: 10 to Severe Storm: 90). Rain slows transit
5. Synthetic Logistics	In-House Synthetic (Simulation Only)	Personnel, Resource Level, Readiness, Time Limit	Models fuel capacity and vehicle readiness. Lower fuel levels create risk
6. Synthetic Intelligence	In-House Synthetic (Simulation Only)	Observation Type, Zone, Confidence, Reliability	Calculates information uncertainty = 1 - (confidence × reliability). High uncertainty increases risk

4. How the Model Arrives at the Shortest & Most Appropriate Route

One of the most important technical achievements of TACTIX AI is its multi-objective routing algorithm. Here is the step-by-step breakdown of how a route is discovered and selected:

Step A: Creating the Unified Environment Grid

The map area is divided into a regular grid of **30 rows × 35 columns = 1,050 cells**. Every individual cell carries precomputed environmental values. For any cell *c*, the platform computes its **Single-Cell Environment Cost (0 to 100)**:

Environment Cost Formula:
Cost(cell) = [0.35 × TerrainScore] + [0.25 × LandCoverFactor × 100] + [0.25 × WeatherImpact] + [0.15 × (1 - RoadAccessibility) × 100]
• Terrain: 35% weight | Land Cover: 25% weight | Weather: 25% weight | Road Access: 15% weight (Sums to 100%).

Step B: Building the 8-Connected Planning Graph

Each cell in the grid represents a **Node**. Each node connects to its 8 surrounding neighbor cells (North, South, East, West, and 4 Diagonals). A step between cell A and cell B represents an **Edge**. If a neighbor is deep water without a bridge, that connection is pruned.

Step C: The A* Search Algorithm in Action

To find a route from Start to Goal, TACTIX AI executes the **A* (A-Star) Pathfinding Algorithm**:

- 1. **gScore(n)**: The exact accumulated cost traveled from the Start cell to the current cell *n*.
- 2. **hScore(n) [Heuristic]**: The estimated Great-Circle (Haversine) physical distance remaining from cell *n* to the Goal cell. Because the heuristic never overestimates physical distance, A* is mathematically guaranteed to find the optimal path.
- 3. **fScore(n) = gScore(n) + hScore(n)**: The total estimated path cost through cell *n*.
- 4. **MinHeap Priority Queue**: At each step, the algorithm pops the cell with the lowest fScore from a binary MinHeap, explores its 8 neighbors, updates their tentative scores, and repeats until the Goal cell is reached.

Step D: Why Three Distinct Simulated Options (COAs) Are Generated

Instead of presenting only one route, TACTIX AI generates **three genuinely distinct Courses of Action (COAs)** by varying the edge-cost objective function over the exact same graph:

Option Name	Optimization Objective	Mathematical Edge-Cost Formula	Behavioral Strategy
■ Option Alpha (Speed Priority)	Minimize simulated duration	Cost(a, b) = Distance(a, b) × 1.0 + [EnvCost(b) / 100] × 0.2 (Roads receive a 40% speed bonus)	Pushes along major highway corridors to achieve the fastest travel time
■ Option Bravo (Resource-Opt)	Minimize fuel burn & vehicle wear	Cost(a, b) = ResourceBurn(a, b) × 1.0 + [EnvCost(b) / 100] × 0.4 (Where ResourceBurn penalizes uphill climbs)	Steers onto paved, level grades with steady speeds to conserve resources
■ Option Charlie (Hazard Bypass)	Minimize environmental hazard exposure	Cost(a, b) = [EnvCost(b) / 100] × 1.0 + Distance(a, b) × 0.3 (Water proximity heavily penalized 3.0×)	Avoids steep slopes, muddy basins, and riverbanks to reduce environmental impact

Step E: Constraint Engine & Rule Validation

Once routes are generated, each route passes through the **Constraint Engine**:

- **Time Limit:** If Route Duration > Scenario Time Limit → Classified as **INVALID (X)**.
- **Resource Limit:** If Fuel Burn > Available Resource Envelope → Classified as **INVALID (X)** (or **WARNING (■)** if within 10%).
- **Restricted Zones:** Ray-casting polygon intersection detects if any route segment enters an ecological sanctuary or hazard corridor → Classified as **INVALID (X)**.
- **Preferred Alternative Selection:** The system highlights the lowest-risk **VALID** option as the '*Preferred Simulated Alternative*' (never labeled as 'best military plan').

5. The Explainable Risk Engine & AI Architecture

The 6-Factor Weighted Risk Formula (0 to 100)

Every Course of Action receives an explainable risk score computed from 6 documented factors:

Factor	Weight	What Drives This Factor?	Formula / Range
Terrain Risk	22% (0.22)	Mean slope gradient and elevation roughness along the route	$\text{avg}(\text{TerrainScore})$ on scale 0–100
Weather Risk	22% (0.22)	Precipitation intensity, storm gusts, and visibility class.	$\text{Weather.impact_score}$ on scale 0–100
Logistics Stress	20% (0.20)	Fuel availability, equipment readiness, supply depletion.	$100 - (\text{ResourceLevel} \times \text{Readiness} \times 100)$
Intel Uncertainty	18% (0.18)	Reliability and sensor gaps in nearby synthetic intelligence	$\text{avg}(\text{Uncertainty})$ of reports within 2.2 km buffer
Time Pressure	10% (0.10)	Proximity of estimated transit duration to the hard deadline	$\text{interp01}(\text{Duration} / \text{TimeLimit} - 0.4) \times 166.7$
Constraint Stress	8% (0.08)	Warnings (18 pts each) and hard rule violations (45 pts each)	$\text{Warnings} \times 18 + \text{Violations} \times 45$ (capped at 100)

How the AI Works (Interpreter & Analyst)

The platform integrates two specialized AI modules via serverless API routes:

- AI Mission Interpreter (POST /api/interpret):** Takes free-text user orders (e.g. *Move through north sector under monsoon rain within 5 hours*) and converts them into structured JSON: { weather: 'HEAVY_RAIN', resource_level: 0.55, time_limit: 5, terrain_preference: 'AVOID_WATER' }. The output is strictly validated against a Zod schema before touching the simulator.
- AI Decision Analyst (POST /api/analyze):** Takes the pre-computed metrics and risk tables and generates plain-language executive summaries, trade-off comparisons, and what-if explanations. The prompt strictly instructs the LLM: *'Use ONLY the numbers provided; never invent metrics; present options as lower-risk simulated alternatives'*.
- Machine Learning Validation:** A Scikit-learn Random Forest model trained on 3,000 scenario perturbations achieved **MAE = 0.296**, **RMSE = 0.388**, **R² = 0.9989**, proving the mathematical consistency of the underlying weighted risk model.

6. End-to-End User Experience & Website Workflow

When an operator uses the TACTIX AI website, they experience an intuitive 4-stage progressive disclosure workflow:

Stage	User Action & Interface Experience	Underlying System Execution
① Describe	User opens the site, completes or skips the 5-step Guided Tour.	The TypeScript frontend dispatches a request to a code interpreter.
② Review	A clean confirmation card appears showing extracted Weather, Resources, Time, and Terrain.	The server initializes the Ed05CeoEnv.
③ Compare	Staged loader displays progress. The Tactical Map renders a path through the objective passes.	The simulation computes the optimal path.
④ Explore	User clicks [Why?] buttons to inspect risk factors.	The simulator's Agentic capabilities respond to the AI Analyst's explanation.

Summary: TACTIX AI bridges the gap between complex GIS spatial science, deterministic simulation algorithms, and accessible human decision-making. By keeping the core simulation deterministic and utilizing AI purely for translation and narrative explanation, the platform delivers zero-latency, highly explainable, and ethically grounded decision support.